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Act-and-wait time-delayed feedback control of autonomous systems

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ABSTRACT

Recently an act-and-wait modification of time-delayed feedback control has been proposed for the stabilization of unstable periodic orbits in nonautonomous dynamical systems (Pyragas and Pyragas, 2016 [30]). The modification implies a periodic switching of the feedback gain and makes the closed-loop system finite-dimensional. Here we extend this modification to autonomous systems. In order to keep constant the phase difference between the controlled orbit and the act-and-wait switching function an additional small-amplitude periodic perturbation is introduced. The algorithm can stabilize periodic orbits with an odd number of real unstable Floquet exponents using a simple single-input single-output constraint control.

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1. Introduction

Time-delayed feedback control (TDFC) [1] is a simple and convenient experimental tool for the stabilization of unstable periodic orbits (UPOs) in nonlinear dynamical systems. The control law of the algorithm does not require any detailed knowledge of the system under control except of the period of the controlled orbit. The stabilization of a UPO is attained by a feedback signal composed from the difference between the current output signal and the output signal delayed by the period of the UPO.

Applications of the TDFC in divers experimental systems as well as theoretical achievements in the field of the TDFC can be found in Refs. [2–5]. To improve the performance of the TDFC a number of modifications have been proposed: extended TDFC (ETDFC) [6], TDFC with variable and distributed delays [7,8], and the TDFC with an adaptive tuning of delay time and coupling strength [9–11].

Many publications considered the so called odd number limitation of TDFC. In Ref. [12], it was stated that the TDFC cannot stabilize UPOs with an odd number of real unstable Floquet exponents (FEs). In order to bypass the limitation various modifications of the TDFC have been developed [13–17]. Later it has been shown that the usual TDFC can overcome the limitation in autonomous systems, but only in the framework of multiple-input multiple-output (MIMO) constraint control [18–22].

The theory of TDFC is difficult because it deals with infinite-dimensional function space. Since the stability of a controlled UPO is determined by an infinite number of Floquet exponents (FEs), the optimization of the control parameters is a severe problem (see e.g. [23]). A very attractive idea that allows to transform the TDFC into a finite-dimensional controller is to apply an act-and-wait concept, which implies a periodic switch on (act) and off (wait) of the delayed feedback term [24–29]. It has been shown that the time-delayed system can be treated as a finite-dimensional system provided that the duration of waiting is longer than the duration of acting.

Recently we managed to implement the act-and-wait idea into TDFC algorithm, but only for the case of nonautonomous systems [30]. For the act-and-wait TDFC, it is necessary to keep constant an appropriate phase difference between the UPO and the act-and-wait switching function. In the nonautonomous case, this is easy to do by using an information of the driving force of the nonautonomous system.

In this letter, we develop an act-and-wait modification of the TDFC for autonomous systems. Unlike to nonautonomous systems, here we do not have any straightforward information on the phase of the target UPO, and in order to lock the phase of the target UPO with that of the act-and-wait switching function, we introduce an additional small-amplitude periodic perturbation. To find the required phase of the periodic perturbation, we appeal to a phase reduction theory [31,32]. We show that this control works for the systems with an odd number limitation even in the case of a single-input single-output (SISO) constraint control. Moreover, since the number of FEs of the controlled UPO remains the same as for the control-free system, an effective minimization of the spectral abscissa can be achieved.

The paper is organized as follows. In Sec. 2 we formulate the problem. Section 3 is devoted to a linear stability analysis of the act-and-wait TDFC without taking into account the additional pe-

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riodic perturbation. In Sec. 4 we derive a phase locking condition that determines the phase difference between the periodic perturbation and the act-and-wait switching function. Numerical demonstrations for a simple model of the normal form of the Hopf bifurcation [18] as well as chaotic Lorenz [33] and Rössler [34] systems are presented in Sec. 5. We finish the paper with the conclusions presented in Sec. 6.

2. Problem formulation

Let us consider an autonomous dynamical system $\dot{\mathbf{x}}(t) = \mathbf{F}(\mathbf{x}(t))$, where $\mathbf{x}(t)$ is an n-dimensional state vector of the system and $\mathbf{F}(\mathbf{x}(t))$ is an n-dimensional vector function. We assume that the system has an unstable τ-periodic orbit $\mathbf{x}(t) = \xi(t) = \xi(t+\tau)$ that satisfies the equation $\dot{\xi}(t) = \mathbf{F}(\xi(t))$. We seek to stabilize this orbit in the framework of the following single-input single-output (SISO) constraint:

$$\dot{\mathbf{x}}(t) = \mathbf{F}(\mathbf{x}(t)) + \mathbf{e}_m u(t), \tag{1a}$$

$$v(t) = h(\mathbf{x}(t)). \tag{1b}$$

Here $\mathbf{e}_m$ is an n-dimensional unity-length vector with the only component m equal to unity and all other components equal to zero. It specifies to which equation of the system the scalar input variable $u(t)$ is connected. Finally, $v(t)$ is a scalar output variable, which is defined by the state variable $\mathbf{x}(t)$ through a function $h(\mathbf{x}(t))$.

We specify an act-and-wait TDFC algorithm as follows:

$$u(t) = u_0(t) + \varepsilon s(t), \tag{2a}$$

$$u_0(t) = -\kappa g(t)[v(t) - v(t-\tau)]. \tag{2b}$$

Here κ is a feedback gain. The main idea of the act-and-wait TDFC is that the TDFC force is multiplied by the 2τ-periodic act-and-wait switching function

$$g(t) = \begin{cases} 0, & \text{if } 0 \le (t \mod 2\tau) < \tau_w, \\ 1, & \text{if } \tau_w \le (t \mod 2\tau) < \tau_w + \tau_a = 2\tau, \end{cases} \tag{3}$$

so that the control force is switched off for a period of length τ_w (wait), and it is switched on for a period of length τ_a (act). The total period of act-and-wait duration is equal to the double period of the orbit, $\tau_a + \tau_w = 2\tau$. Throughout this work we assume that the condition $\tau_a < \tau_w$ is fulfilled. As is shown in Ref. [30], this condition ensures that the act-and-wait TDFC system is always n-dimensional, i.e. the controller does not change the dimensionality of the system. Thus we can integrate this system as a system of ordinary differential equations (ODEs) without invoking sophisticated numerical techniques required for delay differential equations (DDEs).

Unlike in Ref. [30], where the act-and-wait TDFC was considered for nonautonomous systems, here in the control law (2), an additional small periodic perturbation $\varepsilon s(t)$, $\varepsilon \ll 1$, $s(t+T) = s(t)$ with the period $T = 2\tau$ equal to the period of the act-and-wait switching function is introduced. This perturbation is necessary for control of autonomous systems in order to keep constant an appropriate phase difference between the UPO and the act-and-wait switching function. This phase difference plays a crucial role for the success of the act-and-wait TDFC algorithm. Note that nonautonomous systems have an inherent periodic driving force, which can be used for keeping the above phase difference and no additional periodic perturbation in the control law is needed [30].

For autonomous systems, the profile of a periodic signal $s(t)$ can be chosen arbitrarily. Here we use a simple harmonic signal $s(t) = \sin(\Omega t + \varphi_s)$, where $\Omega = 2\pi/T$ and φ_s defines the phase difference between the periodic signal $s(t)$ and the act-and-wait switching function $g(t)$. We fix the phase of the switching function $g(t)$ so that at $t = 0$ it starts from the waiting period. If the phase of the stabilized UPO is locked with the phase of the periodic signal $s(t)$, then by variation of the φ_s we can control the phase difference between the UPO and the switching function $g(t)$. Note that the above described algorithm is slightly invasive due to the presence of the periodic perturbation $\varepsilon s(t)$, however, below we will show that the required phase locking can be achieved with an arbitrary small amplitude ε of the periodic perturbation.

In the next section, we describe a linear stability analysis of the controlled UPO without taking into account the periodic perturbation, i.e., for $\varepsilon = 0$. The aim of this analysis is to find a set of control parameters at which the controlled UPO becomes stable. One of the important parameters responsible for the stability is the phase difference φ between the UPO and the switching function $g(t)$. In Sec. 4 we invoke a phase reduction theory for finding an appropriate value of the phase φ_s which ensures the phase locking of the stabilized UPO with the desired phase difference φ.

3. Linear stability analysis

For $\varepsilon = 0$, the act-and-wait TDFC perturbation (2) is strictly noninvasive; the UPO of the control-free system remains the solution of the closed-loop system. The linear stability of this solution strongly depends on the phase difference φ between the UPO and the act-and-wait switching function. We remind that the phase of the switching function is fixed so that it starts from the waiting period. Then we define the phase φ as follows. First, we choose some initial conditions $\xi(0) = \mathbf{x}_0$ on the UPO and assign them the phase $\varphi = 0$. The UPO corresponding to these initial conditions is defined as $\xi(t)$. Then the UPO shifted by the phase φ with respect to the switching function is $\xi_\varphi(t) = \xi(t + \varphi/\Omega)$. The linear stability of the phase shifted UPO $\xi_\varphi(t)$ is defined by the variational equation

$$\delta\dot{\mathbf{x}}_\varphi(t) = \mathbf{A}_\varphi(t)\delta\mathbf{x}_\varphi(t) + \mathbf{B}_\varphi(t)\delta\mathbf{x}_\varphi(t-\tau), \tag{4}$$

where $\delta\mathbf{x}_\varphi(t) = \mathbf{x}(t) - \xi_\varphi(t)$ is the deviation of the solution from the target UPO, and $\delta\mathbf{x}_\varphi(t-\tau)$ is its τ-delayed version. The T-periodic matrixes $\mathbf{A}_\varphi(t)$ and $\mathbf{B}_\varphi(t)$ are

$$\mathbf{A}_\varphi(t) = \mathbf{J}_\varphi(t) - \kappa g(t)\mathbf{e}_m\mathbf{q}_\varphi^T(t), \tag{5a}$$

$$\mathbf{B}_\varphi(t) = \kappa g(t)\mathbf{e}_m\mathbf{q}_\varphi^T(t). \tag{5b}$$

Here

$$\mathbf{J}_\varphi(t) = [\partial\mathbf{F}(\mathbf{x})/\partial\mathbf{x}]_{\mathbf{x}=\xi_\varphi(t)}$$

is the Jacobian matrix of the control-free system evaluated on the UPO $\xi_\varphi(t)$ and

$$\mathbf{q}_\varphi(t) = [\partial h(\mathbf{x})/\partial\mathbf{x}]_{\mathbf{x}=\xi_\varphi(t)}$$

is a vector derivative of the scalar function $h(\mathbf{x})$ evaluated on the same UPO.

The variational DDE (4) can be solved as a system of ODEs via an algorithm described in Ref. [30]. The main idea behind this algorithm is to write the equations for a fundamental matrix and solve them in the interval $t \in [0, 2\tau]$ in two steps. In the first step, they are solved in the wait interval $t \in [0, \tau_w]$ as an ODE system, since here there are no delay terms. In the second step, the solution is continued to the act interval $t \in [\tau_w, 2\tau]$ again by solving an ODE system. Though here there are delay terms, their values are obtained from the solution of the system in the wait interval provided $\tau_a \le \tau_w$ or equivalently $\tau \le \tau_w$. So, despite the presence

of the delay, the stability of the controlled UPO is determined by the finite number of Floquet exponents (FEs) $\lambda_1, \lambda_2, \ldots, \lambda_n$, where n is the dimension of the control-free systems. This fact gives the main advantage of the act-and-wait modification of the TDFC with respect to the usual TDFC that produces an infinite number of Floquet exponents. Note that for $\varepsilon = 0$, the Floquet spectrum of the act-and-wait TDFC contains the zero FE (let's say $\lambda_1 = 0$) despite the fact that the closed-loop system is not time-invariant due to the presence of the time-dependent function $g(t)$. The zero exponent exists because the UPO $\xi_\varphi(t)$ remains the solution of the closed-loop system for any phase difference φ. It follows that the initial phase difference φ is neutrally stable and this is why we need an additional small periodic perturbation $\varepsilon s(t)$ in a real experiment. By this perturbation we can lock the phase of the orbit with the phase of the function $g(t)$ and avoid an undesired drift of the phase difference φ due to noise or other small external disturbances.

Note that a neutral stability occurs also in the classical TDFC scheme with the constant feedback gain, when $g(t) \equiv 1$ for all t. However, in that case, the transverse stability of the UPO does not depend on the phase of the orbit and we do need to fix its phase. On the contrary, in the act-and-wait control, the transverse stability drastically depends on the phase difference between the UPO and the act-and-wait switching function and we need to maintain this difference at a desired constant value during the experiment.

The spectrum of the FEs $\{\lambda_j\}$, $j = 2, \ldots, n$ derived from the variational Eq. (4) depends on a set of three control parameters $\mathbf{p} = (\kappa, \tau_a, \varphi)$. In the linear stability analysis, we are interested in the computation of the spectral abscissa (the leading real part of FEs)

$$\Lambda(\mathbf{p}) = \max_{j>1}[\mathrm{Re}\lambda_j(\mathbf{p})] \tag{6}$$

as a function of the control vector $\mathbf{p}$. The controlled orbit is transverse stable if $\Lambda(\mathbf{p}) < 0$. The absolute value of the spectral abscissa $|\Lambda(\mathbf{p})|$ defines the characteristic rate with which the small transverse deviations approach the orbit. In Ref. [30], we have shown how to find an optimal set of control parameters $\mathbf{p}$ that minimizes the function $\Lambda(\mathbf{p})$.

4. Phase locking condition

Now we consider an influence of a small periodic perturbation $\varepsilon s(t)$. The aim of this perturbation is to provide a phase locking between the UPO and the switching function $g(t)$ with a desired phase difference φ via an appropriate choice of the phase φ_s of the perturbation. To find the phase locking condition, we appeal to a phase reduction theory [31]. Specifically, we apply recent results of the theory devoted to time-delay systems [32]. Though the theory in Ref. [32] is developed for time-invariant delay systems, it can be adopted in our case with a time-dependent switching function $g(t)$. To this end, we extend the system by an additional variable $\psi(t) = (\Omega t) \bmod (2\pi)$ that defines the phase of the switching function $g(t)$. Then, using the equalities $\dot\psi(t) = \Omega$ and $g(t) = g(\psi(t)/\Omega)$, for the extended vector $\mathbf{y}(t) = [\mathbf{x}^T(t), \psi(t)]^T$ we obtain an equation in the form $\dot{\mathbf{y}}(t) = \mathbf{\Phi}(\mathbf{y}(t), \mathbf{y}(t-\tau)) + \varepsilon\tilde{\mathbf{e}}_m s(t)$ equivalent to that considered in Ref. [32]. Here $\tilde{\mathbf{e}}_m$ is an $n+1$ dimensional unity-length vector with all components equal to zero, except of the mth component, and $\mathbf{\Phi}(\mathbf{y}(t), \mathbf{y}(t-\tau))$ is an appropriate $n+1$ dimensional vector function. Note that the solution $\mathbf{y}(t) = [\xi_\varphi^T(t), \psi(t)]^T$ of the extended system corresponding to the unperturbed ($\varepsilon = 0$) UPO $\xi_\varphi(t)$ is T-periodic due to the T-periodicity of the function $\psi(t)$.

Using the phase reduction theory developed in Ref. [32], we can describe an influence of a small perturbation $\varepsilon s(t)$ in terms of a scalar phase variable $\Theta(t)$ that satisfies a simple one-dimensional equation

$$\dot\Theta(t) = \Omega + \varepsilon \mathbf{z}_\varphi^T(\Theta)\mathbf{e}_m s(t). \tag{7}$$

Here $\mathbf{z}_\varphi(\Theta)$ is a 2π-periodic phase response curve (PRC) and $\Theta(t)$ defines the phase of the stabilized UPO shifted by an amount φ with respect to the switching function. The PRC can be obtained from the equation

$$\dot{\mathbf{z}}_\varphi(t) = -\mathbf{A}_\varphi^T(t)\mathbf{z}_\varphi(t) - \mathbf{B}_\varphi^T(t+\tau)\mathbf{z}_\varphi(t+\tau) \tag{8}$$

with the periodic boundary condition $\mathbf{z}_\varphi(T) = \mathbf{z}_\varphi(0)$. Here the range of the argument of the PRC is assumed to be $[0, T)$ instead of $[0, 2\pi)$. Equation (8) is adjoint to the linearized variational equation (4). It is composed in such a way that the bilinear form

$$\begin{aligned}(\mathbf{z}_\varphi(t), \delta\mathbf{x}_\varphi(t)) = \mathbf{z}_\varphi^T(t)\delta\mathbf{x}_\varphi(t) + \int_{-\tau}^{0} \mathbf{z}_\varphi^T(t+\tau+\theta) \\ \times \mathbf{B}_\varphi(t+\tau+\theta)\delta\mathbf{x}_\varphi(t+\theta)d\theta\end{aligned} \tag{9}$$

holds in time, i.e., is time-independent. Equation (8) represents a difference-differential equation of advanced type and its periodic solution can be found via a backward integration. The numerical algorithm for finding the PRC is described in Appendix. It is worthy of note that the algorithm is based on integration of ODEs rather than DDEs.

Knowing the PRC, the desired phase locking condition can be estimated from Eq. (7). This equation can be further simplified by making use of the small parameter ε. To this end, we introduce a new variable $\phi(t)$ that defines the phase difference between the phase $\Theta(t)$ and the phase $\psi(t) = \Omega t$ of the switching function, $\phi(t) = \Theta(t) - \Omega t$. Then, from Eq. (7), $\phi(t)$ obeys

$$\dot\phi(t) = \varepsilon \mathbf{z}_\varphi^T(\phi + \Omega t)\mathbf{e}_m s(t). \tag{10}$$

As mentioned above, we use the harmonic external perturbation $s(t) = \sin(\Omega t + \varphi_s)$. Now, because the right-hand side of the above equation is $O(\varepsilon)$, $\phi(t)$ is a slowly varying variable, while Ωt varies rapidly. Thus, we may approximate the right-hand side of the above equation by integrating it over one period of the fast external forcing, assuming that $\phi(t)$ does not vary within the period T. By this averaging approximation, we obtain

$$\dot\phi(t) = \frac{\varepsilon}{T}\int_t^{t+T} \mathbf{z}_\varphi^T(\phi + \Omega t')\mathbf{e}_m \sin(\Omega t' + \varphi_s)dt'. \tag{11}$$

Then by changing the integration variable $t' = (\Psi' - \phi)/\Omega$ and taking into account the 2π-periodicity of the PRC, this equation can finally be presented in the form

$$\dot\phi(t) = \varepsilon\Gamma_\varphi(\varphi_s - \phi) \tag{12}$$

with the right-hand side averaged function

$$\Gamma_\varphi(\varphi_s - \phi) = \frac{1}{2\pi}\int_0^{2\pi} \mathbf{z}_\varphi^T(\Psi')\mathbf{e}_m \sin(\Psi' + \varphi_s - \phi)d\Psi'. \tag{13}$$

The phase locking condition with the desired phase difference φ between the UPO and the switching function is fulfilled when Eq. (12) has a stable stationary solution at $\phi = 0$. Thus the phase φ_s of the external perturbation that provides the desired phase locking is defined by the equation

$$\Gamma_\varphi(\varphi_s) = 0. \tag{14}$$

Below we demonstrate the validity of the above theoretical results by several numerical examples.

5. Numerical demonstrations

In this section, we present numerical results for three examples: the normal form of the subcritical Hopf bifurcation [18], the Lorenz system [33] and the Rössler system [34].

5.1. The normal form of the subcritical Hopf bifurcation

The normal form of the subcritical Hopf bifurcation has been intensively studied in the context of the odd number limitation of the TDFC [12–22]. It was the first model on which the authors of Refs. [18,19] showed that the usual TDFC can overcome the odd number limitation in autonomous system. They demonstrated this using a MIMO configuration in the form of a rotating time-delayed feedback. Here we show that the UPO in this model can be stabilized by an act-and-wait TDFC even in the framework of a simple SISO configuration without recourse to a MIMO control.

The normal form of the subcritical Hopf bifurcation is defined by a two-dimensional state vector $\mathbf{x}(t) = [x_1(t), x_2(t)]^T$ that satisfies an equation $\dot{\mathbf{x}}(t) = \mathbf{F}(\mathbf{x}(t))$ with the vector field

$$\mathbf{F}(\mathbf{x}) = \begin{bmatrix} \lambda & -1 \\ 1 & \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + (x_1^2 + x_2^2) \begin{bmatrix} 1 & -\gamma \\ \gamma & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}. \tag{15}$$

Here λ and γ are system parameters, λ determines the distance from the Hopf bifurcation point, and the Hopf frequency is normalized to unity. The parameter γ governs the dependence of the oscillation frequency on the amplitude. For $\lambda < 0$, the bifurcation is subcritical and the system has a stable fixed point $\mathbf{x} = 0$ and a UPO. The advantage of this system is that the UPO is determined analytically, $\xi_\varphi(t) = R[\cos(\omega t + \varphi), \sin(\omega t + \varphi)]^T$, where $\omega = 1 - \gamma\lambda$ is the frequency, $R = \sqrt{-\lambda}$ is the amplitude, and φ is an arbitrary initial phase of the UPO. The UPO has two FEs $\lambda_1 = 0$ and $\lambda_2 = -2\lambda > 0$ and thus satisfies an odd number limitation condition. In the numerical simulations presented below, we take the following values of the parameters: $\lambda = -0.02$ and $\gamma = 1$. Then the period of the UPO is $\tau = 2\pi/\omega \approx 6.1600$.

The results of applying a SISO act-and-wait time-delayed feedback control to this system are presented in Fig. 1. The controlled system is defined by Eq. (1) with the vector-function (15). We chose the input vector $\mathbf{e}_m = \mathbf{e}_2 = (0, 1)^T$ and the output variable $v(t) = h(\mathbf{x}(t)) = x_1(t)$. To find appropriate control parameters $\mathbf{p} = (\kappa, \tau_a, \varphi)$, we first performed a linear stability analysis of the controlled UPO without taking into account the external periodic perturbation ($\varepsilon = 0$). The Floquet exponents of the controlled orbit were found from Eq. (4). Despite the presence of the delay term, the number of the FEs remains two, as for the control-free system, and one of them remains zero, $\lambda_1 = 0$, for any values of the control parameters. The second, nonzero exponent λ_2, defines the spectral abscissa (6), $\Lambda(\mathbf{p}) = \mathrm{Re}\lambda_2(\mathbf{p})$. By using a gradient sampling algorithm as described in Ref. [30], we found an optimal set of the parameters $\mathbf{p} = \mathbf{p}^*$ that minimizes the spectral abscissa. The dependence of the spectral abscissa on the coupling strength κ for the fixed optimal values $\tau_a = \tau_a^* \approx 0.5150 \times \tau$ and $\varphi = \varphi^* \approx 0.3502 \times 2\pi$ is shown in Fig. 1a. At the optimal value $\kappa = \kappa^* \approx 1.2038$, we see a deep minimum that likely indicates a deadbeat.

In the next step, we fixed the control parameters at the optimal values and found the PRC by solving Eq. (8) as described in Appendix. Then we used the phase locking condition (14) and determined the phase $\varphi_s \approx 0{,}07697 \times 2\pi$ of the external perturbation. Finally, with the given values of the parameters, we performed an integration of the nonlinear control system (1). To estimate the quality of the control, we introduce a relative control error

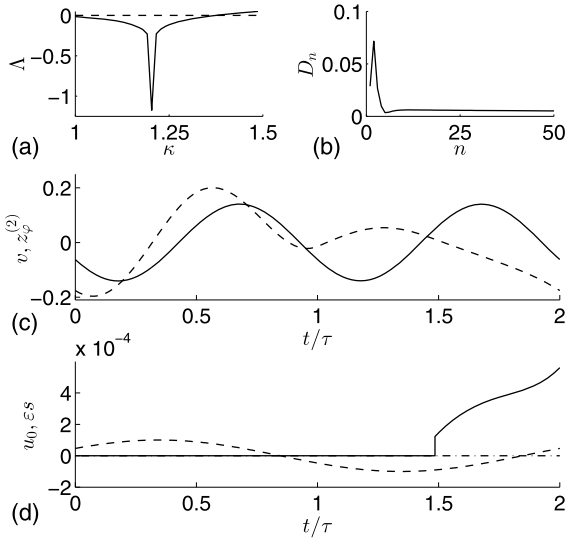


Fig. 1. Act-and-wait time-delayed feedback control of the normal form of a Hopf bifurcation. (a) Spectral abscissa Λ vs. feedback gain κ for the fixed optimal values $\tau_a/\tau \approx 0.5150$ and $\varphi/2\pi \approx 0.3502$. The minimum of Λ is attained at $\kappa \approx 1.2038$. (b) The dynamics of the relative synchronization error D_n defined by Eq. (16). (c) The dynamics of the output variable $v(t) = x_1(t)$ (solid curve) at the 50th period of the act-and-wait control when the transient regime is over and the second component $z_\varphi^{(2)}$ of the PRC (dashed curve). (d) The post-transient dynamics of the perturbations $u_0(t)$ (solid curve) and $\varepsilon s(t)$ (dashed curve) corresponding to the 50th period of the act-and-wait control. The amplitude of the perturbation is $\varepsilon = 10^{-4}$.

$$D_n = \left[\frac{1}{\sigma^2 T} \int_{(n-1)T}^{nT} [v(t) - v(t-\tau)]^2 dt \right]^{1/2}, \tag{16}$$

where $\sigma^2 = (1/\tau) \int_0^\tau [h(\xi(t))]^2 dt$. This quantity measures a relative deviation of the output signal from a τ-periodic solution at the nth period of the act-and-wait control. For strictly noninvasive control, this quantity should vanish as $n \to \infty$. In our case the control is slightly invasive due to the presence of the external perturbation $\varepsilon s(t)$, and in Fig. 1b we see that after a transient regime the quantity D_n converges to a small value. In Fig. 1c, we plot the dynamics of the output variable at the 50th period of the act-and-wait control, when the transient regime is over. This signal converges to the UPO, $v(t) \to h(\xi_\varphi(t)) = \xi_\varphi^{(1)}(t)$, with the desired optimal phase $\varphi = \varphi^*$. Here $\xi_\varphi^{(1)}(t)$ denotes the first component of the UPO with the phase φ. In this figure, we also plot the second component $z_\varphi^{(2)}(t)$ of the PRC, which has been used to estimate the required phase φ_s of the external perturbation. Finally, in Fig. 1d we show the post-transient dynamics of the perturbations $u_0(t)$ and $\varepsilon s(t)$ corresponding to the 50th period of the act-and-wait control. They are of the order of 10^{-4}, which means that the control is only weakly invasive.

In Fig. 2 we show that the asymptotic value $D_{as} = D_{n\to\infty}$ of the control error linearly decreases with the decrease of ε and tends to zero when $\varepsilon \to 0$. Thus, by decreasing the amplitude ε of the external perturbation, we can make the control perturbation arbitrarily small. However, the decrease of ε slows down the phase dynamics and the required phase locking may take a long time.

5.2. The Lorenz system

As the next example satisfying an odd number limitation in autonomous systems, we consider the Lorenz system [33]. The sys-

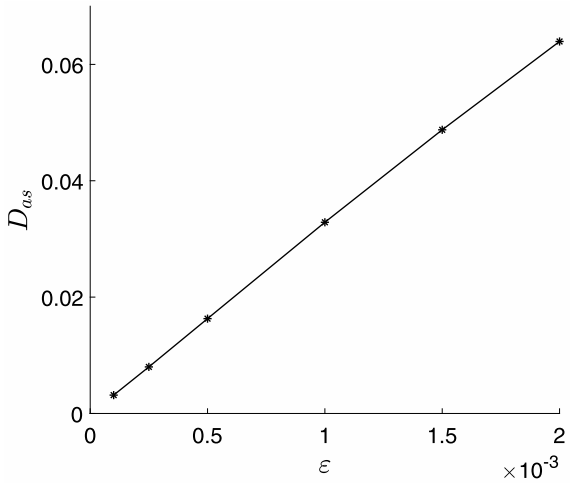


Fig. 2. The dependence of the asymptotic value $D_{as} = D_{n\to\infty}$ of the relative control error on the amplitude ε of the external perturbation for the controlled normal form of the Hopf bifurcation.

tem is defined by a three-dimensional state vector $\mathbf{x} = [x_1, x_2, x_3]^T$ and the vector field

$$\mathbf{F}(\mathbf{x}) \equiv \begin{bmatrix} -\sigma(x_1 - x_2) \\ rx_1 - x_2 - x_1x_3 \\ x_1x_2 - bx_3 \end{bmatrix}. \tag{17}$$

We use the standard values of the parameters $\sigma = 10$, $r = 28$ and $b = 8/3$ that characterize its chaotic behavior. We stress that due to the odd-number limitation the control of the Lorenz system by the TDFC is a nontrivial problem. To do this various modifications such as an unstable TDFC [14] or a half-period TDFC [13] have been devised. The successful control with the usual TDFC in the MIMO configuration has been attained only recently [22].

Here we show that this system can be efficiently controlled by an act-and-wait time-delayed feedback using a simple SISO configuration. We chose an input vector $\mathbf{e}_m = \mathbf{e}_3 = (0, 0, 1)^T$ and an output variable $v(t) = h(\mathbf{x}(t)) = x_3(t)$. As a control target, we chose the period-one UPO with the initial conditions $\xi_0(0) = \mathbf{x}_0 \approx (-15.467, -15.411, 36.598)^T$, and the period $\tau \approx 1.5586$. Performing the procedures similar to those described in the previous example, we achieved the successful control of the UPO. The results are presented in Fig. 3. Here the periodic orbit of the free as well as the controlled system has three FEs. The dependence of the spectral abscissa Λ on the coupling strength κ, shown in Fig. 3a for the values of the parameters $\tau_a = 0.8 \times \tau$ and $\varphi = 0.35 \times 2\pi$, has a minimum at $\kappa \approx 5.25$. For the given values of the parameters, the phase locking condition gives the following value $\varphi_s = 0.1247 \times 2\pi$ of the phase of the external perturbation. Modeling the dynamics of the nonlinear control system (1) with the vector flow (17) shows that it converges to the given UPO with the desired phase φ (Fig. 3c) and the relative control error D_n tends to small values (Fig. 3b). The amplitudes of the control perturbations $u_0(t)$ and $\varepsilon s(t)$ in the post-transient regime are small (Fig. 3d) in comparison to the amplitude of the output variable $v(t)$ (Fig. 3c), which evidences a weak invasiveness of the control.

5.3. The Rössler system

As a last example, we present an act-and-wait TDFC for the Rössler system [34] defined by the vector field

$$\mathbf{F}(\mathbf{x}) = \begin{bmatrix} -x_2 - x_3 \\ x_1 + ax_2 \\ b + x_3(x_1 - c) \end{bmatrix} \tag{18}$$

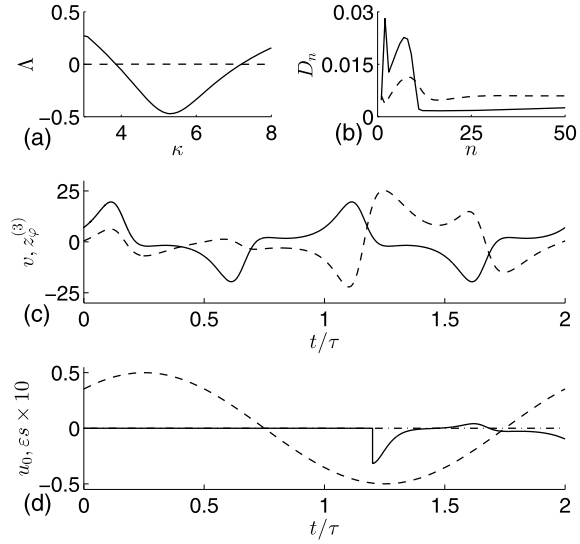


Fig. 3. Act-and-wait time-delayed feedback control of the Lorenz system. The results are presented in the same way as in Fig. 1. An input vector is $\mathbf{e}_m = (0, 0, 1)^T$ and an output variable is $v(t) = x_3(t)$. The values of the parameters are $\varphi/2\pi = 0.35$, $\tau_a/\tau = 0.8$ and $\varepsilon = 0.05$. The minimum of the spectral abscissa Λ is attained at $\kappa = 5.25$. Unlike in Fig. 1, here in (b) we show the dynamics of D_n for two values of ε: 0.05 (solid) and 0.1 (dashed).

with the state vector $\mathbf{x} = [x_1, x_2, x_3]^T$. We use the standard values of the parameters $a = 0.2$, $b = 0.2$ and $c = 5.7$ at which the system's behavior is chaotic. Note that the odd number limitation does not apply to this system, because its UPOs do not have any real positive Floquet exponents and they can be controlled by the usual TDFC algorithm. However, an experience of applying the TDFC to specific systems shows that it is impossible to achieve a deep minimum in the dependence of the spectral abscissa on the coupling strength. A specially developed optimization algorithm for minimizing the spectral abscissa of the MIMO systems with the usual TDFC gives an optimal value of the spectral abscissa that by absolute value is comparable with the leading FE of the uncontrolled UPO [23]. Moreover, highly unstable orbits cannot be stabilized by the usual TDFC or even ETDFC at all. The difficulty of the effective minimization of the spectral abscissa is related to the fact that the periodic orbits under the usual TDFC have an infinite number of FEs, while the number of control parameters is finite [23]. On the contrary, the spectral abscissa can be effectively minimized in the case of the act-and-wait TDFC, since the number of FEs of the controlled orbit remains the same as for the control-free system.

Here we consider the control of the period-one UPO of the Rössler system with the initial conditions $\xi_0(0) = \mathbf{x}_0 \approx (-6.559, -3.099, 0.016)^T$ and the period $\tau \approx 5.881$. In Fig. 4 we show the results of an act-and-wait TDFC in the SISO configuration with an input vector $\mathbf{e}_m = \mathbf{e}_2 = (0, 1, 0)^T$ and an output variable $v(t) = h(\mathbf{x}(t)) = x_2(t)$. The values of the parameters are presented in the figure caption. The main advantage of applying the act-and-wait TDFC to the Rössler system is that the spectral abscissa shown in Fig. 4a has a very deep minimum $\Lambda \approx -0.98$ at $\kappa \approx 0.777$, whose absolute value is larger than the corresponding value of the control-free system ($\Lambda \approx 0.15$ for $\kappa = 0$) almost by an order of magnitude.

6. Conclusions

We have developed the act-and-wait modification of the TDFC algorithm for control of UPOs in autonomous dynamical systems. The modification makes the closed-loop system finite-dimensional despite the presence of the time-delayed feedback. The number of

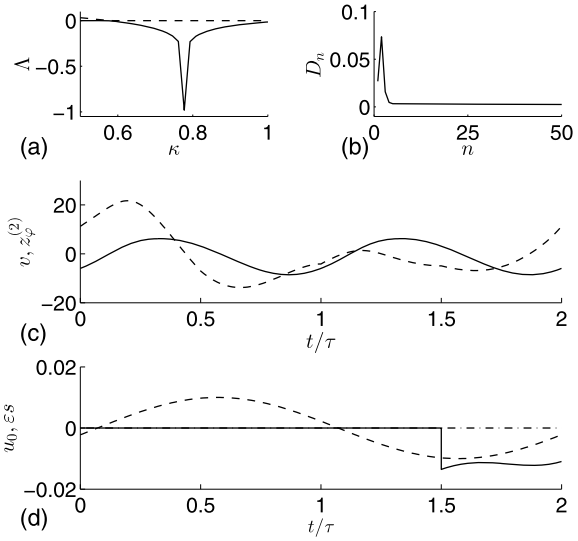


Fig. 4. Act-and-wait time-delayed feedback control of the Rössler system. The results are presented in the same way as in Figs. 1 and 3. An input vector is $\mathbf{e}_m = (0,1,0)^T$ and an output variable is $v(t) = x_2(t)$. The values of the parameters are $\varphi/2\pi = 0.35$, $\tau_a/\tau = 0.5$ and $\varepsilon = 0.01$. The phase of the periodic perturbation is $\varphi_s = 0.9642 \times 2\pi$. The minimum of the spectral abscissa Λ is attained at $\kappa = 0.777$.

Floquet exponents of the controlled UPO becomes equal to that of the control-free system, and they now can be easily manipulated by varying the control parameters. This allows us to attain a deep minimum in the spectral abscissa of the controlled orbit.

One of the important control parameters is the phase difference between the target orbit and the act-and-wait switching function. In order to keep this difference at an appropriate constant value, we introduced an additional small-amplitude periodic perturbation with the period equal to the double period of the target UPO. Using a phase reduction theory, we have derived a phase locking condition that determines the required phase of the perturbation.

The efficacy of the proposed algorithm has been demonstrated with three autonomous systems, namely, the normal form of the subcritical Hopf bifurcation, the Lorenz system and the Rössler system. The UPOs of the first two systems satisfy an odd number limitation and they are difficult to control by the usual TDFC algorithm. The successful control of such UPOs with the usual TDFC can be attained only in a MIMO configuration. Here we have shown that these systems can be effectively controlled by an act-and-wait TDFC even in a simple SISO configuration. In all examples, the successful control was achieved with very small amplitudes of the external periodic perturbation, thus the proposed act-and-wait TDFC is only slightly invasive.

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Appendix A. Algorithm for finding the PRC

The algorithm of finding the PRC is based on a backward integration of the adjoint Eq. (8), similarly as described in Ref. [32]. However, unlike in Ref. [32], here such an integration can be performed with the ODE rather than the DDE integrator. First, we rewrite the Eq. (8), and add the same equation, delayed by τ:

$$\dot{\mathbf{z}}_\varphi(t) = -\mathbf{A}_\varphi^T(t)\mathbf{z}_\varphi(t) - \mathbf{B}_\varphi^T(t+\tau)\tilde{\mathbf{z}}_\varphi(t) \tag{A.1a}$$

$$\dot{\tilde{\mathbf{z}}}_\varphi(t) = -\mathbf{A}_\varphi^T(t-\tau)\tilde{\mathbf{z}}_\varphi(t) - \mathbf{B}_\varphi^T(t)\mathbf{z}_\varphi(t) \tag{A.1b}$$

Here we denoted the delayed version of the PRC as $\tilde{\mathbf{z}}_\varphi(t) \equiv \mathbf{z}_\varphi(t-\tau)$, i.e., the tilde means the τ-delay of the corresponding variable. In all examples presented in this paper, the vector $\mathbf{q}$ is time-independent. Thus the product $\mathbf{e}_m\mathbf{q}^T \equiv \mathbf{K}$ in Eqs. (5), which we denote as a coupling matrix $\mathbf{K}$, is also time-independent. Inserting the expressions $\mathbf{A}_\varphi(t)$ and $\mathbf{B}_\varphi(t)$ from Eqs. (5) into Eqs. (A.1) and noting that $g(t) \equiv 0$ for $t \in [0,\tau]$ and $g(t+\tau) \equiv 0$ for $t \in [\tau, 2\tau]$, we can simplify Eqs. (A.1) in the above time intervals. Specifically, in the time interval $t \in [0,\tau]$ the Eqs. (A.1) read as:

$$\dot{\mathbf{z}}_\varphi(t) = -\mathbf{J}_\varphi^T(t)\mathbf{z}_\varphi(t) + \kappa\, g(t+\tau)\mathbf{K}^T\tilde{\mathbf{z}}_\varphi(t) \tag{A.2a}$$

$$\dot{\tilde{\mathbf{z}}}_\varphi(t) = -[\mathbf{J}_\varphi(t) - \kappa\, g(t+\tau)\mathbf{K}]^T\tilde{\mathbf{z}}_\varphi(t) \tag{A.2b}$$

and in the time interval $t \in [\tau, 2\tau]$ they are:

$$\dot{\mathbf{z}}_\varphi(t) = -[\mathbf{J}_\varphi(t) - \kappa\, g(t)\mathbf{K}]^T\mathbf{z}_\varphi(t) \tag{A.3a}$$

$$\dot{\tilde{\mathbf{z}}}_\varphi(t) = -\mathbf{J}_\varphi^T(t)\tilde{\mathbf{z}}_\varphi(t) + \kappa\, g(t)\mathbf{K}^T\mathbf{z}_\varphi(t) \tag{A.3b}$$

Now we see that the Eq. (A.2b) is actually an ODE that can be solved independently from Eq. (A.2a). Then the Eq. (A.2a) can also be interpreted as a non-homogeneous ODE with the known function $\tilde{\mathbf{z}}_\varphi(t)$ determined from solution of Eq. (A.2b). Similar properties hold for Eqs. (A.3) and they can also be solved as ODEs. We need the solution of Eqs. (A.2) and (A.3) that satisfies the boundary conditions

$$\mathbf{z}_\varphi(2\tau) = \mathbf{z}_\varphi(0) = \tilde{\mathbf{z}}_\varphi(\tau),\ \tilde{\mathbf{z}}_\varphi(2\tau) = \tilde{\mathbf{z}}_\varphi(0) = \mathbf{z}_\varphi(\tau). \tag{A.4}$$

The direct solution of the above boundary problem is problematic, since Eqs. (A.2) and (A.3) are unstable and their solutions grow in time. More appropriate is their backward integration. Then the solution automatically converges to the desired periodic solution. To perform the backward integration, we invert the time in these equations and introduce the new variables $\mathbf{w}(t) = \mathbf{z}_\varphi(2\tau - t)$ and $\tilde{\mathbf{w}}(t) = \tilde{\mathbf{z}}_\varphi(2\tau - t)$. Then, for the time interval $t \in [0,\tau]$, Eqs. (A.3) transform to:

$$\dot{\mathbf{w}}(t) = [\mathbf{J}_\varphi(2\tau - t) - \kappa\, g(2\tau - t)\mathbf{K}]^T\mathbf{w}(t) \tag{A.5a}$$

$$\dot{\tilde{\mathbf{w}}}(t) = \mathbf{J}_\varphi^T(2\tau - t)\tilde{\mathbf{w}}(t) - \kappa\, g(2\tau - t)\mathbf{K}^T\mathbf{w}(t) \tag{A.5b}$$

and, for the time interval $t \in [\tau, 2\tau]$, Eqs. (A.2) read as:

$$\dot{\mathbf{w}}(t) = \mathbf{J}_\varphi^T(2\tau - t)\mathbf{w}(t) - \kappa\, g(\tau - t)\mathbf{K}^T\tilde{\mathbf{w}}(t) \tag{A.6a}$$

$$\dot{\tilde{\mathbf{w}}}(t) = [\mathbf{J}_\varphi(2\tau - t) - \kappa\, g(\tau - t)\mathbf{K}]^T\tilde{\mathbf{w}}(t) \tag{A.6b}$$

Now we outline the algorithm of the backward integration.

(i) Choose a tolerance $\epsilon << 1$, and assign an arbitrary initial condition $\mathbf{w}_0$ for Eq. (A.5a).
(ii) Integrate Eq. (A.5a) with the initial condition $\mathbf{w}(0) = \mathbf{w}_0$ in the time interval $[0,\tau]$. Denote $\mathbf{w}(\tau) = \mathbf{w}_1$.
(iii) Integrate the system of Eqs. (A.5a) and (A.5b) in the time interval $[0,\tau]$ with the initial conditions $\mathbf{w}(0) = \mathbf{w}_0$, $\tilde{\mathbf{w}}(0) = \mathbf{w}_1$. Denote $\mathbf{w}(\tau) = \mathbf{w}_1'$ and $\tilde{\mathbf{w}}(\tau) = \mathbf{w}_0'$.
(iv) Integrate Eq. (A.6b) in the time interval $[\tau, 2\tau]$ with the initial condition $\tilde{\mathbf{w}}(\tau) = \mathbf{w}_0'$. Denote $\tilde{\mathbf{w}}(2\tau) = \mathbf{w}_1''$.
(v) Integrate the system of Eqs. (A.6) in the time interval $[\tau, 2\tau]$ with the initial conditions $\mathbf{w}(\tau) = \mathbf{w}_1''$, $\tilde{\mathbf{w}}(\tau) = \mathbf{w}_0'$. Denote $\mathbf{w}(2\tau) = \mathbf{w}_0''$ and $\tilde{\mathbf{w}}(2\tau) = \mathbf{w}_1'''$.
(vi) If $|\mathbf{w}_0'' - \mathbf{w}_0| < \epsilon$ then go to the step (vii). Otherwise assign $\mathbf{w}_0 = \mathbf{w}_0''$ and go to the step (ii).
(vii) Flip the time series of $\mathbf{w}(t)$ in the interval $[0, 2\tau]$ in order to obtain the solution in the original time direction: $\mathbf{z}_\varphi(t) = \mathbf{w}(2\tau - t)$.

For the examples presented in this paper, the backward integration converges to the desired periodic solution with the tolerance $\epsilon \sim 10^{-7}$ already after about 10 periods of UPO. Note for our problem we do not need the knowledge of the absolute value of the norm of the PRC, since the phase locking condition (14) does not depend on this norm. However, Eq. (14) has both stable and unstable phase locking solutions for φ_s. As is seen from Eqs. (12) and (13), the stability of the solution depends on the sign of the PRC. From Eq. (13) it follows that the change of the PRC's sign is equivalent to the shift of the phase φ_s by π. Thus the stable and unstable phase locking solutions φ_s of Eq. (14) differ by the value π. Which of the two solutions is stable we verify by direct numerical integration of the system (1).

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